Teddy Bear Wheelchair Individual Prototype Report

Section and Team Number: 8

Author (Name, ID): Deep Patel (1379878)

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Statement:

This report was completed by the author alone, in collaboration with my group members, as part of my TBWC project requirements.

Name: ___Deep Patel___

Signature: _ ___

1. Introduction

Throughout the course of the Teddy Bear Wheelchair (TBWC) project, my role changed from primarily focusing on design aesthetics to being involved in nearly every subsystem of the prototype. Even though my allotted category was the design and aesthetics component with Humza, I ended up getting deeply involved in the construction, assembly, and detailing of both rolling chassis and launcher mechanisms. I was the central person involved in the building and development of the physical structure of the wheelchair, making sure that ideas were translated well into a working prototype.

Other than actual work, I was also a team leader in coordination and organization. I structured our meeting times to accommodate everyone's schedule so that we could consistently meet and get things done. Our team met every Tuesday after 5 o'clock. and on Thursdays after 6 o'clock, with additional sessions whenever required on other days. I made sure work was being accomplished effectively and supported my peers where assistance was required. We had Humza and me on the design aesthetic side, Sami and Ayaan in the electronic section, and Ryan and Brendon on the chassis and launcher side. But on observing discrepancies in workload allocation and major accountability gaps, I stepped in to help enforce leadership of the group's progression and section interaction. Through coordinating leadership functions with technical hands-on input, I was able to assist design improvement as well as the successful integration of our TBWC prototype.

2. Contribution Elements

Complete the table below to identify the details of your contributions to the project to date and how they fit into the design process or are considered as additional contributions. Guidelines for your entries are listed below. Refer to the document “ENGG*1100 F25 Prototype Report Rubric – Individual” on CourseLink for further guidance.

- Write 1-3 sentences in each box for each phase of the design process you contributed to. Note: you likely won't have contributed to each phase, or every aspect.
- Some examples of design process phases are: Empathize with people, define the problem, generate ideas, prototype, testing solutions, assess results.
- Be sure to reference your Logbook as evidence of your contribution

Project Aspect	Design Project Phase and Contribution.
Subsystem 1–Rolling Chassis	I contributed extensively on the creation of the rolling chassis from design idea generation to the final testing. I have designed and developed a number of layout concepts, basing designs from team brainstorming and eventually helping settle on the three-wheel rectangular configuration to provide more stability and allow for the motor and launcher. I built nearly the entire base frame with Sami, assembling the frame, securing the wheel axles, and mounting the motor with proper alignment and balance. I also helped set the electronic components in a convenient position on the chassis for easy access and weight balance. When the motor did not function during testing, I helped in identifying the malfunction, which was a faulty battery casing, and supplying a replacement to retest. Because I attended each meeting and saw developments firsthand, I could integrate new ideas as they evolved, oversee the overall construction process, and make certain the chassis evolved in a progressive manner into a stable and reliable foundation for the entire TBWC prototype. (Reference Logbook Group Meetup 4 through Group Meetup 7)
Subsystem 2 – Launching Mechanism	I created the first concept for the mousetrap-powered launcher because I realized it was the most efficient and smallest power source for delivering consistent launch motion. Though Brendan was willing to construct the mechanism, his design was too heavy, limiting power output and reducing range. Using the same design idea for the core design, I proceeded to rebuild and improve

	the launcher, reducing its weight considerably without sacrificing structural integrity. The reduced weight design worked much better, having more powerful and consistent launches. I also determined how to install the launcher on the chassis for best balance and tip avoidance during fire. Apart from this, I did some design drawings of the launcher and team and solo testing to ascertain that improvements in performance are achieved. (Reference Logbook entries: Group Meetup 4 through Solo Work Session 2, and Lab 4.)
Aesthetics & Accessibility	Empathize: I was a part of the aesthetics team and wanted to ensure our TBWC design not only looked nice but was accessible in reality as well. I considered how usability, safety, and design simplicity could work hand in hand to create an effective and approachable prototype. I understood how real wheelchairs optimize user stability, evenly distributed weight, and shielding of electrical parts, and integrated these into our teddy bear wheelchair. I minimized the overall design to a rectangular shape with more balance. This required also considering the axle length as well as readjusting the three-wheel arrangement to compensate for the shorter secondary axle. Define and Ideate: During the design process, I integrated safety and functionality in every structural consideration, so that the wiring could be housed safely, the chassis would offer mounting locations for both the launcher and electronics, and the weight would be evenly loaded to prevent tipping. I researched how the placement of front and rear axles would affect maneuverability, and how the added bumper would increase ground clearance and shock absorption. I also considered ease of access to internal components in order to guarantee that maintenance or adjustment could be done easily without the need to open the entire prototype. Prototype and Refine: Throughout the process, I

	collaborated very closely with other chassis and electronics group teammates to apply these design decisions into the actual build. My goal was to ensure the prototype not just functioned but was also safe, durable, and neatly arranged with thoughtful aesthetics in consideration. I continuously modified the visual and structural design as additional issues arose, with prime concern given for a clean, useful, and properly balanced design that would demonstrate technical reliance as well as aesthetics. (Refer to Logbook entries: Group Meetup 2 through Group Meetup 3.)
Team Contract Document	Throughout the project timeframe, the original team contract was good for task division and expectations but, as the designing process went on, our duties varied widely from what had been initially set out. Though my role fell under the design aesthetics, I was mainly involved in nearly all aspects of the project, including the construction of the rolling chassis and launcher systems. Most of us naturally gravitated to work in spaces where our strengths were most needed, overlapping tasks and collaborative problem-solving rather than strictly assigned work. For example, while Sami and Ayaan were originally assigned electronics, I helped design the layout and placement of those items to create stability and accessibility. Likewise, while Brendan was developing the launcher, I contributed significantly to the construction and development of those systems. Despite all these transitions, the basic values that we had outlined in our team agreement, communication, responsibility, and accountability to one another mostly continued to be central to our process. I ensured I created meeting schedules, managed schedules, and ensured that team members worked on their parts during meetings. Our team sometimes struggled with consistency in attendance and responsibility, therefore we made up by reassigning tasks and constantly updating work sessions.

	(Reference S7 - 8: Week 4 - TBWC - Section 07 - Team Contract 1)
Additional Contributions	In addition to my design and technical contribution, team and project management responsibilities were mine. Regular scheduling of meeting times and booking meeting rooms to guarantee efficient use of our time were ensured by me. I set firm goals and priorities prior to each meeting on what was to be done, and, in the course of meetings, I assigned personal tasks to guide everyone and keep them on track. I also monitored progress and checked in with group members to make sure tasks got accomplished prior to the next session. Where there were problems or delays, I helped redistribute assignments and relaunch focus so the group could stay on track. I made sure communication between all subgroups, design, electronics, and chassis remained well and everyone understood how their tasks contributed to the overall prototype. I encouraged collaboration and helped facilitate minor disputes involving design approaches or timelines to maintain our team environment respectful and effective. Through organization, goal setting, and encouraging responsibility, I ensured the team was well-structured and cohesive throughout the project timeline.